

MANOJ KUMAR CHATTI

Tampa, FL-33647 | +1 6562056856 | manojkchatti@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Results-driven Software Engineer with 3 years of experience building scalable systems, microservices, and high-performance APIs, optimizing distributed systems for low latency. Skilled in AWS, Docker, CI/CD, and data engineering, with experience in AI-powered RAG systems. Adept in Agile/Scrum, delivering reliable and scalable solutions.

PROFESSIONAL EXPERIENCE

Environmental Protection Commission

Aug 2025 - Present

Software Engineer Intern

Tampa, FL

- Developed internal engineering tools and scalable UI modules using React and Node.js, increasing team delivery efficiency by 20%.
- Designed secure backend services with Express and FastAPI, integrating JWT-based authentication and access control, cutting manual request handling by 30%.
- Refactored legacy Java-based systems into Dockerized microservices deployed on AWS ECS, improving service responsiveness by 25%.

Fat Dog Spirits

May 2025 - Aug 2025

Software Engineer Intern

Tampa, FL

- Created two customer-facing dashboards (React + Flask), integrating REDCap survey exports with SQL/NoSQL stores for real-time insights.
- Containerized development setups using Docker, significantly reducing environment configuration issues for new contributors.

Cognizant

Mar 2022 - Aug 2024

Software Engineer

Mumbai, India

- Transformed legacy platforms (Struts) into scalable React + TypeScript and Spring Boot microservices, reducing page latency by seconds and enabling reuse across multiple product teams.
- Architected and delivered 12+ production-grade REST/GraphQL APIs using Java 11/17 and Spring Boot, supporting 5,000+ daily transactions with sub-100 ms latency across distributed banking systems.
- Orchestrated event-driven microservices using AWS SNS/SQS, enabling asynchronous workflows, improving fault isolation and simplifying cross-service recovery mechanisms.
- Engineered resilient systems with idempotency, timeout strategies, and standardized error contracts, strengthening service stability and enabling seamless downstream integrations.
- Automated and scaled CI/CD pipelines using Jenkins, Git, Maven, and Terraform, cutting 8+ hours of weekly manual effort and increasing deployment reliability and consistency.

PROJECTS

CiteRAG Production Retrieval-Augmented Generation System | [Link](#)

- Spearheaded a production-grade RAG system using FastAPI, LangChain, and FAISS over 5K+ documents, delivering source-grounded responses through a containerized (Docker) full-stack setup (React + FastAPI).
- Enhanced retrieval quality using MMR search and cross-encoder reranking, improving relevance and reducing hallucinations by 35%.
- Optimized system performance with Redis caching and API rate limiting, cutting latency by 40% and improving observability.

TECHNICAL SKILLS

- Languages:** Java, Python, JavaScript, TypeScript, SQL, Bash
- Frontend:** React.js, HTML5, CSS3, Dash, REST API Integration
- Backend & Frameworks:** Node.js, Express.js, FastAPI, Flask, Spring Boot, GraphQL
- Databases & Data Platforms:** PostgreSQL, Oracle, Snowflake, BigQuery, Amazon Redshift, NoSQL
- Cloud & DevOps:** AWS (ECS, SNS, SQS), Docker, Terraform, GitHub Actions, Jenkins, CI/CD Pipelines
- Data Engineering & Analytics:** ETL Pipelines, Databricks, Redis, Tableau, Power BI, Data Modeling
- Machine Learning & AI:** LangChain, FAISS, RAG Systems, Cross-Encoder Reranking, MMR Search
- Software Engineering Concepts:** Microservices Architecture, RESTful API Design, Event-Driven Systems, Distributed Systems, Multithreading, System Design, Authentication & Authorization (JWT), Caching, Performance Optimization
- Methodologies:** Agile, Scrum, Software Development Lifecycle (SDLC)

EDUCATION

University of South Florida

Master of Science, Computer Science

Aug 2024 - May 2026

Tampa, FL, USA